

Fault Lines & Flywheels

Seven notes from an uneasy summer - on money, machines, magnets and the small frictions that become a world order.

THE SHORT VERSION

Markets are being tugged by two contrary forces. Fault lines are the hidden stresses: costly debt, rising yields, conflict and brittle supply chains. Flywheels are the engines that keep gathering speed: AI, chips, robotics, infrastructure and national industrial ambition. The interesting bit is that they are happening at the same time - which is rarely boring, and seldom cheap.

\$1tn

AI capex expected in 2027,
including neo-clouds

99%

China share of heavy
rare-earth refining

\$4.7tn

Japan foreign portfolio
assets

-25.3%

IBM intraday drop after its
Q2 warning

This is not a forecast dressed up as certainty. It is a collection of short observations from the supplied research - where the ground is solid, where it is soft, and where it may be worth keeping your shoes on.

Source trail: Eight supplied research papers dated 6-20 July 2026. Figures are reported as sourced, not independently refreshed.

Contents

These pieces can be read in any order. They are connected by a single question: what happens when the world builds faster than it can comfortably finance?

01	The bond market has started knocking on the door Debt, Japan and the quiet power of a higher borrowing cost.
02	AI is a feast. The bill has not arrived yet. Infrastructure booms, memory shortages and an awkward return-on-investment question.
03	Cheaper brains, stubborn real life Why better AI mathematics may matter as much as additional chips.
04	The race is moving from the cloud to the factory floor China, Japan and the return of industrial ambition.
05	The empire of magnets Rare earths and the strategic importance of very small, very dull things.
06	The price of oil is not the price of disruption How maps, straits and freight rates reach the supermarket shelf.
07	A pocket guide for the next few months The hinges to watch when the larger door begins to move.

The source notes at the end name every supplied paper. The prose is interpretive; the uncertainty is deliberate.

The bond market has started knocking on the door

Money used to be quiet. Borrow it cheaply, roll it over, and let tomorrow worry about tomorrow. Tomorrow has now arrived, wearing sensible shoes and carrying an invoice.

The recurring worry in the 13D and Russell Napier material is that debt is no longer a background condition. It is becoming the weather. When government borrowing costs rise, the effect spreads: mortgages, company loans, public budgets and the prices investors are willing to pay for future profits all feel the pressure.

Japan is the laboratory in this story. The sources describe a country with a weak yen, rising government-bond yields and a vast pool of savings that policymakers may try to keep at home. That is not a small domestic matter. Japan is reported to have roughly **\$4.7 trillion** in foreign portfolio assets. If even a part of that money turns around, markets elsewhere notice.

France is the most exposed character in this particular drama, according to the supplied analysis. It has high debt, political uncertainty and a bond market that relies heavily on foreign investors. Bond markets do not generally shout. They simply charge more. The shouting comes later.

"A bond yield is just a price. Until it becomes a verdict."

The number that makes this feel less abstract

354% is the cited Japanese total non-financial debt-to-GDP ratio. Meanwhile, the French 10-year bond yield premium over Germany was reported at **81 basis points** on 17 July. These are gauges, not prophecy; they tell you the engine is running hot.

Source trail: Russell Napier, Japan Inc. is Back; 13D, What I Learned This Week; Jefferies, Asia Maxima.

AI is a feast. The bill has not arrived yet.

The AI boom resembles a very expensive banquet. The memory makers are already serving dessert. The hyperscalers are still working out whether anyone will pay for the main course.

Jefferies estimates that the four largest US cloud companies could spend around **\$700 billion** on capital equipment this year. Add Oracle, OpenAI, Anthropic and the newer cloud players, and the 2027 number rises above **\$1 trillion**. That is approximately 3% of US GDP - too large to call a niche technology trend with a straight face.

For now, the money has been clearest for the people selling the picks and shovels: memory, servers, storage, chips and the plumbing around them. The three largest DRAM makers are reported to control 89% of market revenue. In a shortage, the toll booth earns before the tourist reaches the view.

The uncomfortable question is return on investment. The big cloud companies are projected to devote 92% of operating cash flow to capex in 2026, up from 41% in 2023, and some have increased borrowing too. That does not mean the story ends badly. It means the story now needs an ending other than "build more."

A useful distinction

AI demand can be real while AI economics are still unresolved. A data center can be fully booked and an investment can still be too expensive. Both statements can coexist, like two people who insist they are "fine" at a dinner party.

IBM offered a miniature version of this shift. Its early Q2 warning said customers were directing budgets into servers, storage and memory before software and consulting. IBM fell 25.3% intraday. Yet the supplied data still points to growth in overall technology spending. The point is not "software is over." The point is "the wallet has moved."

Source trail: Jefferies, Asia Maxima - AI crescendo; IBM sector shock deep dive.

Cheaper brains, stubborn real life

The most overlooked part of the AI story may be this: better mathematics can be worth as much as more machinery. A clever shortcut can make a crowded server room feel suddenly bigger.

The CLSA research describes two ways of squeezing more from existing hardware. MiniMax's sparse-attention method claims to cut attention compute by 28.4 times at very long context lengths. DeepSeek's speculative-inference method claims 60-85% faster generation for a user at the same throughput. In plain English: do not read every word in the library if the answer is probably in the last chapter, and do not make the large model do work a smaller model can confidently draft.

That is good news for AI adoption and a small warning for anyone treating scarcity as an eternal law. Memory and compute can stay tight for years, but software has a habit of finding side doors. The history of technology is full of bottlenecks that looked permanent right before somebody found a better wrench.

There is a useful dose of humility here. The "Agents' Last Exam" benchmark found the hardest real professional tasks had an average full-pass rate below 1%, even for leading systems. The machines may be learning to reason. They are still rather bad at understanding why the client quietly changed their mind after page 17.

"A model can remember a million tokens. It still cannot always read the room."

Why this matters beyond technology

If models become cheaper and more capable, AI spreads further into the economy. But the last mile remains human: judgment, context, responsibility and knowing when not to send the email.

Source trail: CLSA, Alpha abstract: Heaven & Earth; 13D, What I Learned This Week.

The race is moving from the cloud to the factory floor

AI used to be discussed as though it lived in a glowing rectangle. The more strategic version lives in ports, warehouses, cars, machine tools and robots that can trip over a cable in real life.

China's push is broad and practical. The supplied 13D research describes a 2 trillion RMB data-center programme using at least 80% domestic hardware, while Chinese models are reported to be getting closer in capability and cheaper in operation. The important ambition is not merely to win a benchmark. It is to wire AI through factories and daily industry.

Japan is pursuing a related but distinctly Japanese project: physical AI. The supplied Japan x NVIDIA brief describes Noetra, a consortium built around a domestic AI foundation model for machines, supported by reported public funding of up to **1 trillion yen** over five years. It sits on top of Japan's industrial robotics base and a stated goal of 10 million AI-equipped robots by 2040.

The philosophical point is oddly old-fashioned. Countries are rediscovering that intelligence is not only something you own in a server. It is something you embed in the tools that make, move and repair the physical world. The future may not be an app. It may be a forklift with excellent manners.

The stack in one sentence

Data -> models -> chips -> robots -> factories. The country that can keep this loop close to home has more control over its industrial future. That is the logic of sovereignty in the supplied material.

Source trail: Japan x NVIDIA Physical AI Brief; 13D, What I Learned This Week; Jefferies, Asia Maxima.

The empire of magnets

Some of the world's most consequential objects are small, dull and nearly invisible. A heavy rare-earth magnet is one of them. Nobody writes poetry about dysprosium - until they cannot build a motor without it.

The 13D research says China refines 99% of global heavy rare-earth supply and produces 94% of permanent magnets. These materials sit inside the less glamorous machinery of modern life: defense systems, electric vehicles, industrial robots, wind turbines, electronics and parts of the semiconductor supply chain.

The implication is not that every mineral company is a treasure chest. It is that a supply chain can be strategically vital long before the end customer notices it exists. The research estimates \$6.5 trillion of downstream production outside China could be exposed under full Chinese export restrictions. Whether that precise estimate proves right or not, the direction of travel is plain: provenance now matters.

This is where the future gets bureaucratic. Governments want proof of where minerals came from, how they were processed and whose rules were followed. Somewhere, an accountant is becoming a national-security asset. History has a sense of humor.

"In a world of strategic scarcity, the boring parts become the crown jewels."

A price signal worth watching

The supplied research says European dysprosium prices rose from around \$240/kg before Chinese controls to \$700-1,000/kg, while terbium rose from \$900/kg to \$2,000-4,000/kg. These are stress signals in an emerging two-tier market.

Source trail: 13D, What I Learned This Week.

The price of oil is not the price of disruption

A barrel of crude is easy to chart. The cost of a delayed ship, a scarce diesel shipment, a disrupted grain route and a nervous insurer is harder to squeeze into a neat line. Unfortunately, households buy the second set of things.

The Jefferies geopolitical material treats the Strait of Hormuz, the Red Sea and the Black Sea as connected pressure points. The message is not simply that oil might rise. It is that freight, refined fuels, industrial inputs and food can all become more expensive even when the quoted oil price looks relatively calm.

Russia's reported refinery disruption is a useful example. The material says crude-processing rates had fallen to their lowest level since 2005, while diesel costs can travel through freight, agriculture, construction and mining. Inflation is often a relay race: the baton changes hands several times before it reaches the supermarket shelf.

For markets, this is a reminder that the central-bank story cannot be read only from a CPI print. Cooler headline inflation can be real and still be fragile if energy and trade routes are one bad week away from making the next reading less friendly.

What to watch

Safe shipping through chokepoints, fuel and freight costs, insurance availability, and whether higher input costs are bleeding into goods prices. War has many terrible consequences; an invoice is among the quieter ones.

Source trail: Jefferies, GREED & fear: Traps and the escalation dynamic; 13D, What I Learned This Week.

A pocket guide for the next few months

There are too many moving parts to reduce this moment to one grand call. A more useful habit is to watch the hinges: the points where a small change can swing a larger door.

WATCH THIS	WHY IT MATTERS	THE HUMAN VERSION
US 10-year yield	Asia Maxima highlights 4.5% as a warning level and 5% as a clearer risk signal.	Higher borrowing costs make the future feel more expensive.
Japanese capital flows	Domestic repatriation could affect bonds and equities far beyond Japan.	When a giant saver comes home early, the party notices.
AI cash returns	Paid usage and cash flow need to catch up with data-center spending.	The feast is lovely; someone must eventually settle the bill.
Rare-earth access	More controls or price gaps would test industrial supply chains.	The tiny magnet becomes the boss of the giant factory.
Freight and diesel	They transmit conflict into everyday prices faster than many investors expect.	A map on the wall becomes a receipt in a pocket.

The central tension remains elegant and awkward: **the same world that is building extraordinary industrial capacity is also becoming more financially brittle**. The right question is not whether everything is wonderful or terrible. It is where the spending is essential, where the supply is scarce, and where the optimism is being financed by someone else's future cash flow.

"The future is arriving as both a factory and a bill."

A final note on tone

The supplied research contains analysis, forecasts and some unverified claims. This guide preserves that uncertainty on purpose. A good map should show the cliffs, not pretend they are flowerbeds.

Source notes

This report synthesises the eight supplied documents. It does not independently verify their figures, dates, interpretations or source quality. Several documents are explicitly opinion-led, scenario-based or based on supplied source material.

Documents used

1. 13D Research & Strategy, *What Are the Markets Telling Us?*, 19 July 2026.
2. Russell Napier / Orlock Advisors, *Japan Inc. is Back*, 20 July 2026.
3. 13D Research & Strategy, *What I Learned This Week*, 16 July 2026.
4. Jefferies, *GREED & fear: Traps and the escalation dynamic*, 16 July 2026.
5. Jefferies, *Asia Maxima - AI crescendo*, 6 July 2026.
6. CLSA, *Alpha abstract: Heaven & Earth*, 10 July 2026.
7. *IBM sector shock deep dive*, 14 July 2026.
8. *Japan x NVIDIA Physical AI Brief*, 16 July 2026.

Not investment advice

This is a reading guide, not a recommendation to buy, sell or hold anything. It does not take account of individual objectives, circumstances or appetite for risk - including the risk of reading a market newsletter just before bed.

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